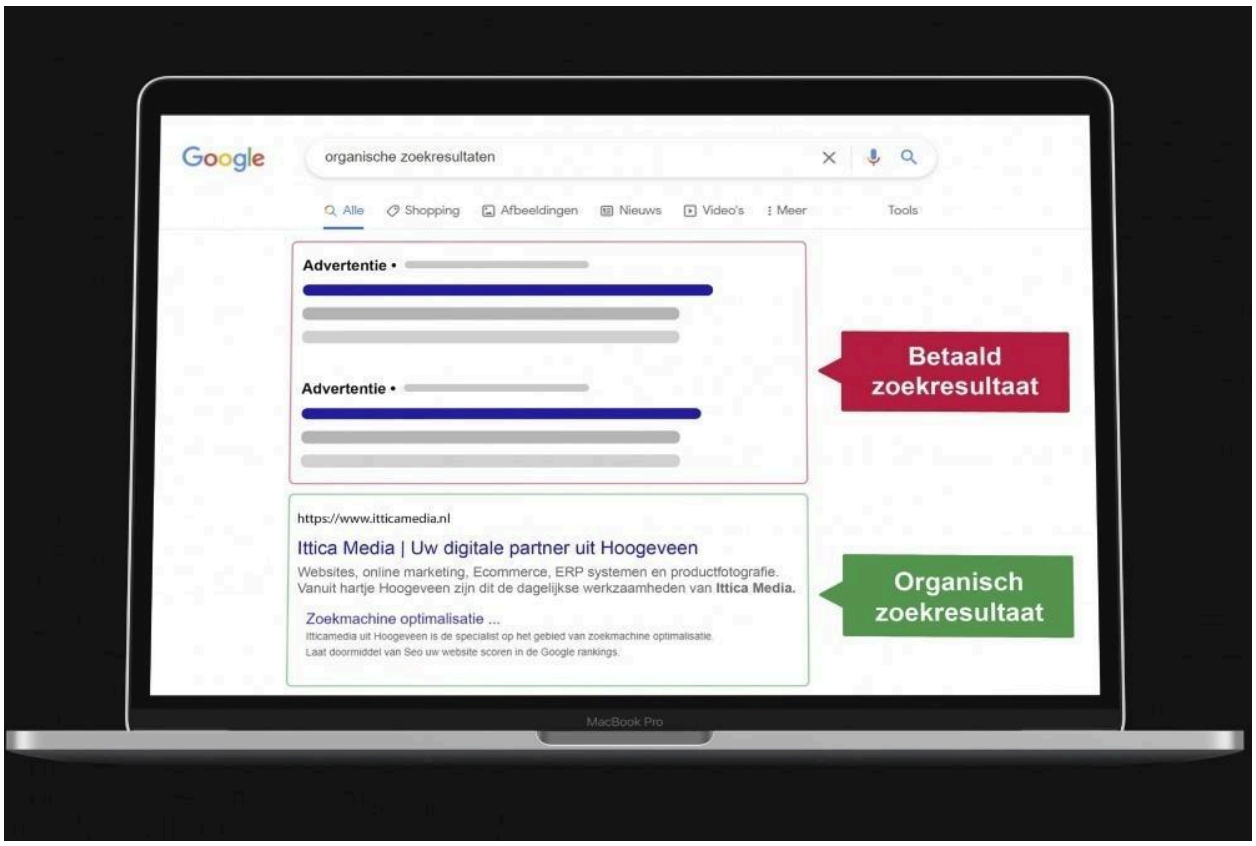
Practical Assignment 4: Search Engine Ranking and Evaluation



Application: Build a mini search engine for a college library.

Problem statement: Develop a system that searches a collection of college library documents and ranks them according to their relevance to a student's query.

Tasks to perform

1. Prepare a document collection: Create at least 10 sample documents containing book titles, descriptions, or research paper abstracts.
2. Implement ranking models:
    - Vector Space Model to represent documents as vectors.
    - TF-IDF to calculate the importance of query terms.
    - Probabilistic Retrieval Model to estimate document relevance.
    - Binary Independence Model (BIM) using binary term presence or absence.

    - Explain how the Probabilistic Ranking Principle (PRP) can guide ranking decisions.
3. Rank the results: Enter a query such as “machine learning applications in healthcare” and display the top five documents.
4. Evaluate the results: Ask a teacher or a group of students to mark documents as relevant or non-relevant. Calculate Precision, Recall, F-measure, Mean Reciprocal Rank (MRR), Mean Average Precision (MAP), and NDCG.
5. Compare the models: Identify which model returns the most useful results and explain why.

Example dataset

Document	Content	Relevant to query?
D1	Machine learning in medical diagnosis	Yes
D2	Healthcare data analysis using AI	Yes
D3	History of computer networks	No
D4	Deep learning for disease prediction	Yes
D5	Database management systems	No

Query: machine learning in healthcare

Students must calculate ranking scores and evaluate the returned results using the relevance labels.

Expected output

- Ranked search results.
- Comparison of at least two ranking models.
- Evaluation metric calculations with interpretation.

- A short report explaining which ranking model performs better.

Concepts covered: Vector Space Model, TF-IDF, Probabilistic Retrieval Model, PRP, BIM, Precision, Recall, F-measure, MRR, MAP, NDCG, test collections, and relevance judgments.